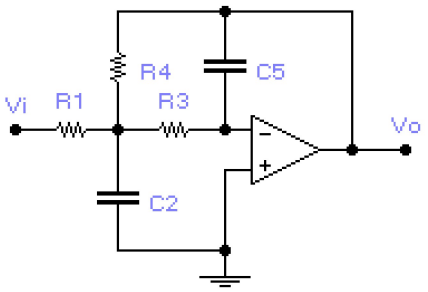
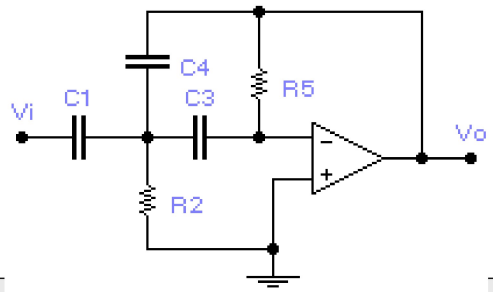
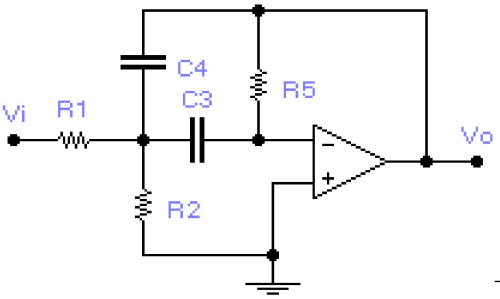
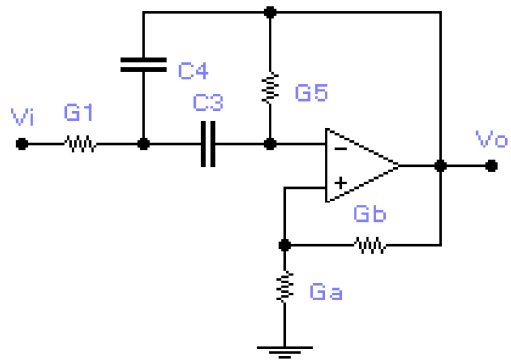
Filtros Activos MFB (Multiple Feedback)

Pasa Bajos	
Red Circuital	***Función de Transferencia***
	$$H(s)=\frac{-G_1G_3}{s^2C_2C_5+sC_5(G_1+G_3+G_4)+G_3G_4}$$ *Ecuaciones de Diseño (Normalizadas):* $G_1=\frac{H}{\omega_0}$; $C_2=\frac{Q(2\omega_0^2+H)}{\omega_0^2}$; $G_3=\omega_0$ $G_4=G_3$; $C_5=\frac{\omega_0^2}{Q(2\omega_0^2+H)}$
Pasa Altos	
Red Circuital	***Función de Transferencia***
	$$H(s)=\frac{-s^2C_1C_3}{s^2C_3C_4+sG_5(C_1+C_3+C_4)+G_2G_5}$$ *Ecuaciones de Diseño (Normalizadas):* $C_1=H$; $G_2=\omega_0(2+H)Q$; $C_3=1$ $C_4=C_3$; $G_5=\frac{\omega_0}{Q(2+H)}$
Pasa Banda	
Red Circuital	***Función de Transferencia***
	$$H(s)=\frac{-sG_1C_3}{s^2C_3C_4+sG_5(C_3+C_4)+G_5(G_1+G_2)}$$ *Ecuaciones de Diseño (Normalizadas):* $G_1=H$; $G_2=2\omega_0Q-H$; $C_3=1$ $C_4=C_3$; $G_5=\frac{\omega_0}{2Q}$

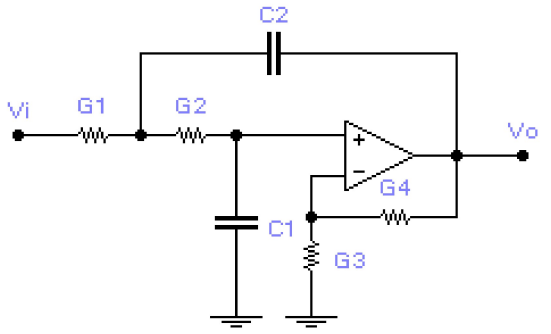
Filtro Activo Pasa Banda MMFB (Modified Multiple Feedback)

Red Circuital	***Función de Transferencia***
	$$H(s)=\frac{-sC_3G_1(1+K)}{s^2C_3C_4+s[G_5(C_3+C_4)-KC_3G_1]+G_1G_5}$$ *Ecuaciones de Diseño:* $\omega_0=\sqrt{\frac{G_1G_5}{C_3C_4}}$; $Q=\frac{\sqrt{\frac{G_1}{G_5}}}{\sqrt{\frac{C_3}{C_4}}+\sqrt{\frac{C_4}{C_3}}-K\frac{G_1}{G_5}\sqrt{\frac{C_3}{C_4}}}$ $K=\frac{G_b}{G_a}$. *Generalmente* $C_3=C_4$

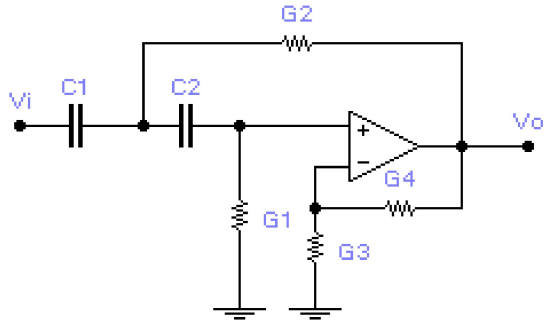
Dr. Ing. Franco Martín Pessana
fpessana@frba.utn.edu.ar

Filtros Activos VCVS (Voltage Controled Voltage Source)

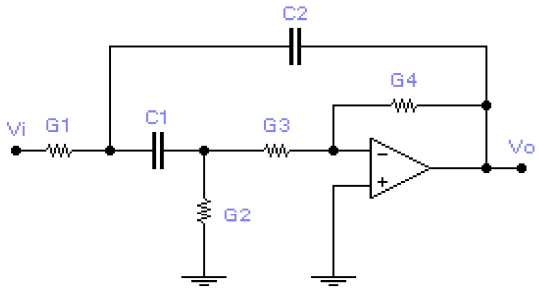
Pasa Bajos

Red Circuital	*Función de Transferencia*
	$H(s)=\dfrac{G_1G_2(1+K)}{s^2C_1C_2+s[(G_1+G_2)C_1-KG_2C_2]+G_1G_2}$ *Ecuaciones de Diseño:* $K=\dfrac{R_4}{R_3}\;;\quad \omega_0=\dfrac{1}{\sqrt{R_1R_2C_1C_2}}$ $Q=\dfrac{1}{\sqrt{\dfrac{R_2C_1}{R_1C_2}}+\sqrt{\dfrac{R_1C_1}{R_2C_2}}-K\sqrt{\dfrac{R_1C_2}{R_2C_1}}}$ *Generalmente* $R_1=R_2$

Pasa Altos

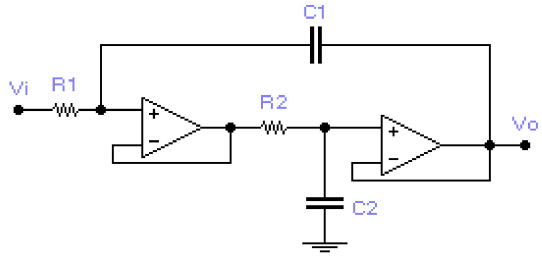
Red Circuital	*Función de Transferencia*
	$H(s)=\dfrac{C_1C_2(1+K)s^2}{s^2C_1C_2+s[(C_1+C_2)G_1-KG_2C_2]+G_1G_2}$ *Ecuaciones de Diseño:* $K=\dfrac{R_4}{R_3}\;;\quad \omega_0=\dfrac{1}{\sqrt{R_1R_2C_1C_2}}$ $Q=\dfrac{1}{\sqrt{\dfrac{R_2C_2}{R_1C_1}}+\sqrt{\dfrac{R_2C_1}{R_1C_2}}-K\sqrt{\dfrac{R_1C_2}{R_2C_1}}}$ *Generalmente* $R_1=R_2$

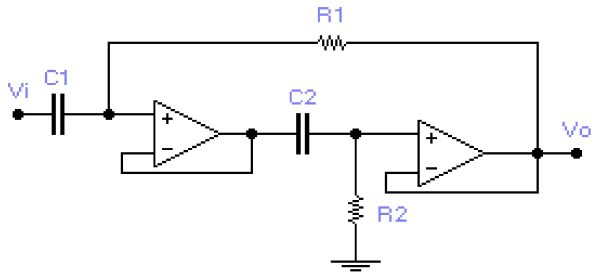
Pasa Banda

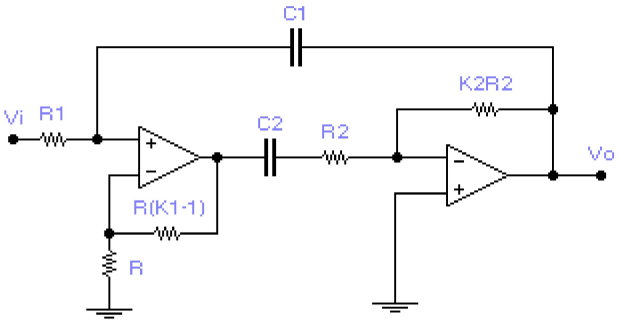
Red Circuital	*Función de Transferencia*
	$H(s)=\dfrac{-KG_1C_1s}{s^2C_1C_2(1+K)+s[G_1C_1+G_2(C_1+C_2)]+G_1G_2}$ *Ecuaciones de Diseño:* $K=\dfrac{R_4}{R_3}\;;\quad \omega_0=\dfrac{1}{\sqrt{R_1R_2C_1C_2(1+K)}}$ $Q=\dfrac{\sqrt{1+K}}{\sqrt{\dfrac{R_2C_1}{R_1C_2}}+\sqrt{\dfrac{R_1C_1}{R_2C_2}}+\sqrt{\dfrac{R_1C_2}{R_2C_1}}}$ *Generalmente* $R_1=R_2$

Dr. Ing. Franco Martín Pessana
fpessana@frba.utn.edu.ar

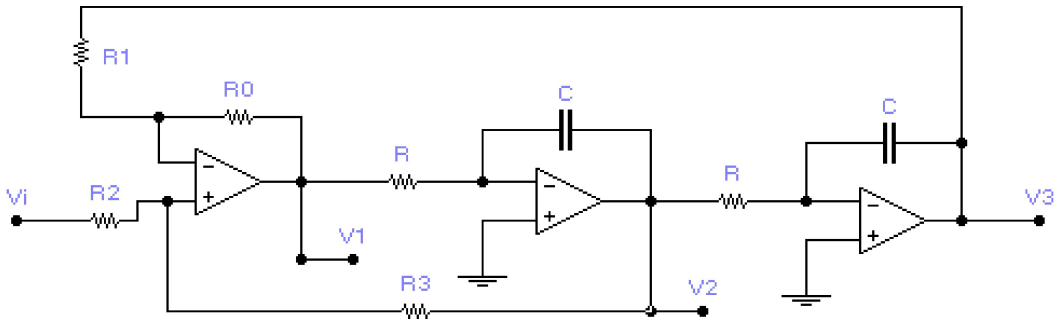
Filtros Activos Bicuadradas

Pasa Bajos	
Red Circuital	***Función de Transferencia***
	$H(s)=\dfrac{1}{s^2T_1T_2+sT_2+1}$ *Ecuaciones de Diseño:* $T_1=R_1C_1\;;\quad T_2=R_2C_2$ $\omega_0=\dfrac{1}{\sqrt{T_1T_2}}\;;\quad Q=\sqrt{\dfrac{T_1}{T_2}}$

Pasa Altos	
Red Circuital	***Función de Transferencia***
	$H(s)=\dfrac{s^2T_1T_2}{s^2T_1T_2+sT_1+1}$ *Ecuaciones de Diseño:* $T_1=R_1C_1\;;\quad T_2=R_2C_2$ $\omega_0=\dfrac{1}{\sqrt{T_1T_2}}\;;\quad Q=\sqrt{\dfrac{T_2}{T_1}}$

Pasa Banda	
Red Circuital	***Función de Transferencia***
	$H(s)=\dfrac{-sK_1K_2T_2}{s^2T_1T_2(1+K_1K_2)+s(T_1+T_2)+1}$ *Ecuaciones de Diseño:* $T_1=R_1C_1\;;\quad T_2=R_2C_2$ $\omega_0=\dfrac{1}{\sqrt{T_1T_2(1+K_1K_2)}}\;;\quad Q=\dfrac{\sqrt{T_1T_2(1+K_1K_2)}}{T_1+T_2}$

Celda Activa Multifiltro



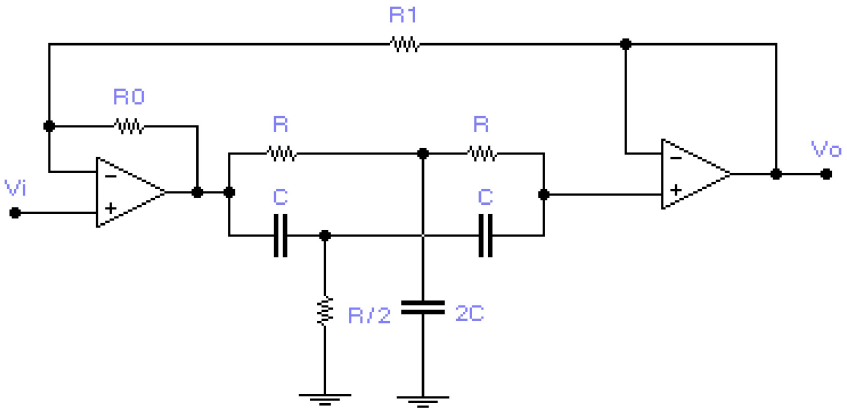
Función de Transferencia

$$\frac{V_3(s)}{V_i(s)} = \frac{K_2}{s^2T^2 + sK_3T + K_1}; \qquad \frac{V_2(s)}{V_i(s)} = \frac{-sK_2T}{s^2T^2 + sK_3T + K_1}; \qquad \frac{V_1(s)}{V_i(s)} = \frac{s^2T^2K_2}{s^2T^2 + sK_3T + K_1}$$

Siendo:

$$T = RC; \qquad K_1 = \frac{R_0}{R_1}; \qquad K_2 = \frac{R_3}{R_2 + R_3}\left(1 + \frac{R_0}{R_1}\right); \qquad K_3 = \frac{R_2}{R_2 + R_3}\left(1 + \frac{R_0}{R_1}\right)$$

Filtro Notch



Función de Transferencia

$$\frac{V_0(s)}{V_i(s)} = \frac{s^2T^2 + 1}{s^2T^2 + s\dfrac{4T}{1+K} + 1}$$

<u>Ecuaciones de diseño</u>:

$$\omega_0 = \frac{1}{T} = \frac{1}{RC}; \qquad Q = \frac{1+K}{4}; \qquad K = \frac{R_0}{R_1}.$$ *Generalmente se elige C.*

Dr. Ing. Franco Martín Pessana
fpessana@frba.utn.edu.ar